

Topic 1.2 - Rates of Change

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

Rates of change turn covariation into a measurable comparison. The same quotient can represent speed, price per item, temperature change per hour, or another context-specific quantity. A correct rate must be attached to its interval and interpreted with units.

Learning targets

- Calculate and interpret average rates of change from formulas, tables, graphs, and contexts.
- Compare rates on different intervals and estimate a rate at a point using nearby average rates.
- Simplify difference quotients and explain what the algebra reveals about changing rates.

Language and structure

Term	Meaning in this topic
average rate of change	$\frac{f(b)-f(a)}{b-a}$ over $[a, b]$, with output units per input unit.
secant line	A line through two graph points; its slope equals average rate of change.
difference quotient	An expression such as $\frac{f(a+h)-f(a)}{h}$, where $h \neq 0$.
local rate estimate	A short-interval secant rate used to estimate behavior near an input.
net change	The output difference $f(b) - f(a)$, before division by input change.

Launch - reason before calculating

A tank drops from 800 L to 620 L during the first 3 hours, then from 620 L to 500 L during the next hour. Compare the two average rates and explain why comparing only the two volume decreases is misleading.

Concept 1: Average rate, interval, and units

Core idea

For distinct inputs a, b , use $\frac{f(b)-f(a)}{b-a}$. Keep subtraction order consistent. The sign describes net direction, while magnitude describes output change per input unit. Contextual interpretation must name both quantities and units.

Worked example - annotate the reasoning

Problem. Customers are modeled by $C(t) = 120 + 18t - 2t^2$, where t is hours after opening. Find and interpret the average rate on $[1, 4]$.

Model reasoning. $C(1) = 136, C(4) = 160$, so the rate is $24/3 = 8$ customers per hour. The customer count increased by an average of 8 customers for each hour over the interval.

Check your understanding

Independent

No calculator

For $H(t) = 4t^2 - 3t + 10$, calculate the average rate on $[2, 5]$.

Concept 2: Tables, nonuniform inputs, and nearby intervals

Core idea

When input steps differ, divide each output change by its own input change. To estimate a rate at $x = c$, use a short interval near or around c and state that the result is an estimate. Very short intervals may magnify rounding or measurement error.

Worked example - annotate the reasoning

Problem. A sensor reports $P(3.8) = 41.2$ and $P(4.2) = 42.0$. Estimate the rate at $x = 4$.

Model reasoning. The symmetric estimate is $(42.0 - 41.2)/(4.2 - 3.8) = 2$ output units per input unit.

Check your understanding

Independent

No calculator

Given $Q(1.9) = 12.44$ and $Q(2.1) = 13.08$, estimate the rate at $x = 2$.

Concept 3: Symbolic difference quotients

Core idea

Expand $f(a+h)$ carefully, subtract the whole expression $f(a)$, factor out h , and only then cancel under the condition $h \neq 0$. The simplified expression shows how an interval rate depends on the starting input and interval length.

Worked example - annotate the reasoning

Problem. For $f(x) = x^2 - 5x + 2$, simplify $\frac{f(a+h)-f(a)}{h}$.

Model reasoning. The numerator is $2ah + h^2 - 5h = h(2a + h - 5)$, so the quotient is $2a + h - 5$.

Check your understanding

Independent

No calculator

For $g(x) = 3x^2 + 2x - 1$, simplify $\frac{g(a+h)-g(a)}{h}$.

Representation laboratory

Connect at least two representations

A graph passes through $(0, 4)$, $(2, 10)$, $(5, 7)$, $(9, 19)$. Compute the consecutive displayed average rates, describe one possible graph, and state what the four points do not determine.

Challenge laboratory

Challenge - justify every claim

Construct $f(x) = ax^2 + bx + c$ with $f(0) = 1$, average rate 5 on $[0, 2]$, and average rate 9 on $[2, 4]$. Verify all conditions.

Error clinic and calculator strategy

Common errors to diagnose

- Using net change as the rate without dividing by input change.
- Using a common denominator for table intervals of different lengths.
- Reporting a rate without units or interval.
- Claiming an exact point rate from a long secant interval.

Technology decision

Technology can evaluate models and compare short intervals efficiently. Keep enough precision, record the interval, and distinguish a decimal estimate from an exact conclusion. A calculator does not supply the contextual interpretation.

AP-style synthesis

Multi-part reasoning task

Water depth $D(t)$, in centimeters, is recorded at selected times.

t (min)	0	3	7	8	12
$D(t)$	18	27	35	33	21

- A. Calculate and interpret the average rate on $[0, 7]$.

- B. Compare the rates on $[3, 7]$ and $[7, 8]$ and interpret their signs.

- C. Explain why the table does not determine the exact time of greatest depth.

Exit ticket

Before you leave

1. State the average-rate formula and units.

2. Why must $h \neq 0$ in a difference quotient?

3. Why is a nearby secant rate usually called an estimate at a point?
